

TEAM KINYETI

STAKEHOLDER ENGAGEMENT PLAN FOR 'OPTIMIZING RAG DOCUMENT RETRIEVAL FOR AGRONOMIC ADVICE' PROJECT

This plan identifies everyone with a stake in Team Kinyeti's retrieval-augmented generation (RAG) system for agronomic advice, and lays out how the team will engage each group throughout the project. The goal is to build a tool that genuinely helps extension officers and farmers find trustworthy agricultural information faster, without losing sight of the people affected by how it is built, tested, and used. This plan is grounded in the values already stated in the project's problem statement: accessibility, transparency, accuracy, and inclusivity.

Primary Stakeholders

Smallholder farmers are the ultimate beneficiaries, though they do not interact with the system directly since it is English-only and mediated through extension officers. Engagement with this group means understanding what questions they actually ask, in what language and phrasing, so retrieval reflects real-world queries rather than assumptions.

Agricultural extension officers are the group that matters most. They query the system, interpret results, and translate or explain them to farmers. Engagement with officers focuses on validating that retrieved documents are useful in the field and not just high-scoring on metrics like nDCG@5, confirming that source traceability is clear enough for them to trust and explain results, and flagging cases where a retrieved excerpt could be misapplied, such as advice meant for a different season or soil type.

Rural advisory programs are kept informed on system capabilities and limitations so they can set realistic expectations with the officers and communities they support.

Secondary Stakeholders

IITA and NAERLS are the source organizations behind the project's document collection. They are consulted on appropriate use and attribution of their materials, and on known gaps or regional limitations in their content. NAERLS in particular focuses on Nigeria, which already limits how well the system generalizes to other African contexts.

Competition organizers and evaluators receive updates per the competition's submission and reporting schedule, since their feedback shapes technical direction, including retrieval method comparisons and evaluation metrics.

Tertiary Stakeholders

Ministry of Agriculture and other policy bodies represent a longer-term engagement, mainly relevant if the project moves toward pilot deployment. Non-English-speaking farming communities are indirectly affected but not directly served, since the dataset and system are English-only; this limitation is documented transparently rather than implied away. Over time, as the project grows, it could evolve to cover non-English speakers as well.

Engagement Approach

Extension officers are engaged through user interviews, scenario walkthroughs, and sample query testing, particularly before major retrieval changes and at key project milestones. Farmer needs are gathered indirectly, relayed through extension officers on an ongoing basis. IITA and NAERLS are contacted as needed, especially before any public claims are made about data coverage. Team members coordinate continuously through standups, shared documentation, and code review. Competition organizers receive submission updates and progress reports on the competition's own schedule, and advisory programs receive a briefing document summarizing system capabilities and limitations at project milestones.

Key Benefits

Engaging these stakeholders directly improves the quality and trustworthiness of the system. Extension officers gain a faster way to locate reliable agronomic guidance instead of searching manually through long manuals and reports, which matters most in rural areas where trained agronomists are scarce. Farmers benefit indirectly through faster, more accurate answers reaching them via officers. IITA and NAERLS benefit from their existing knowledge base being put to practical use, with proper attribution and traceability preserved. Rural advisory programs and policy bodies gain a clearer picture of what the system can and cannot do, which supports responsible adoption rather than overreliance. Keeping these groups engaged throughout the project also means problems, such as retrieval mistakes or regional limitations, surface early rather than after deployment.

Key Risks

Extension officers could over-trust retrieved excerpts as final answers rather than as material to interpret, so onboarding material will clearly position officers as interpreters and not the system as an authority. The system is likely to perform better on Nigerian data from NAERLS than elsewhere, so external communications will be explicit about this limitation rather than overstating generalizability. Because farmers never see the system directly, their real needs could get lost without a structured feedback channel through extension officers. Data usage terms with IITA and NAERLS will be confirmed early rather than after public submission, and the current exclusion of non-English speakers will be documented transparently rather than implying broader accessibility than the system delivers today.

Feedback Loop

Officer feedback on retrieval quality and usability is collected through scenario testing and informal interviews. Recurring issues, such as irrelevant results, missing context, or translation gaps, are logged and routed either to the retrieval and reranking pipeline for technical iteration or to documentation and onboarding materials for trust and usability issues. This plan itself is revisited at each major project milestone to confirm it still reflects who is actually affected by the system.

This plan should evolve as the project moves from competition submission toward any real-world piloting. The current scope treats extension officers as the primary active stakeholder, with farmers

engaged indirectly through them.